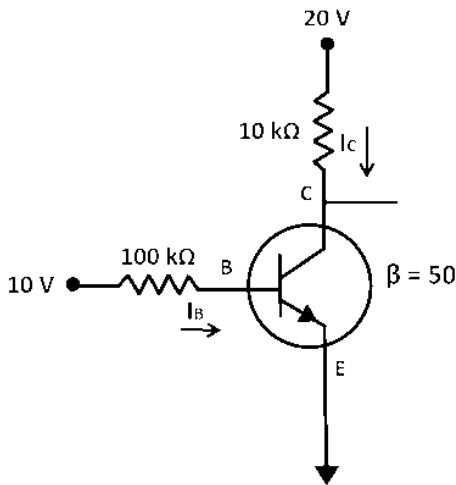


BJT Basics

Review Exercise 1



Assuming the given BJT is in forward active mode-

- Find I_B .
- Find I_C .
- Find V_C .
- Verify if the BJT is in forward active mode.

Ans: a) 0.093 mA
b) 4.65 mA
c) -26.5 V
d) $V_{CE} > 0.2$

Review Practice Problem 1

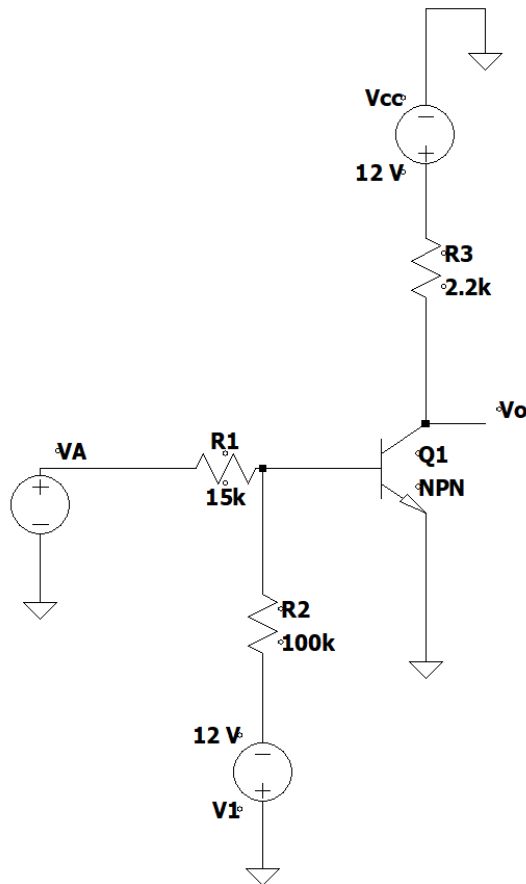
If the BJT in **Review Exercise 1** is in saturation mode-

- Find I_B .
- Find I_C .
- Find I_E .

Ans: a) 0.092 mA
b) 1.98 mA
c) 2.072 mA

Basic Operation

Exercise 2



For the RTL inverter, find the maximum & average power dissipation.

[High input = 12 V, Low input = 0.2 V]

Ans: 74.856 mW, 38.0746 mW

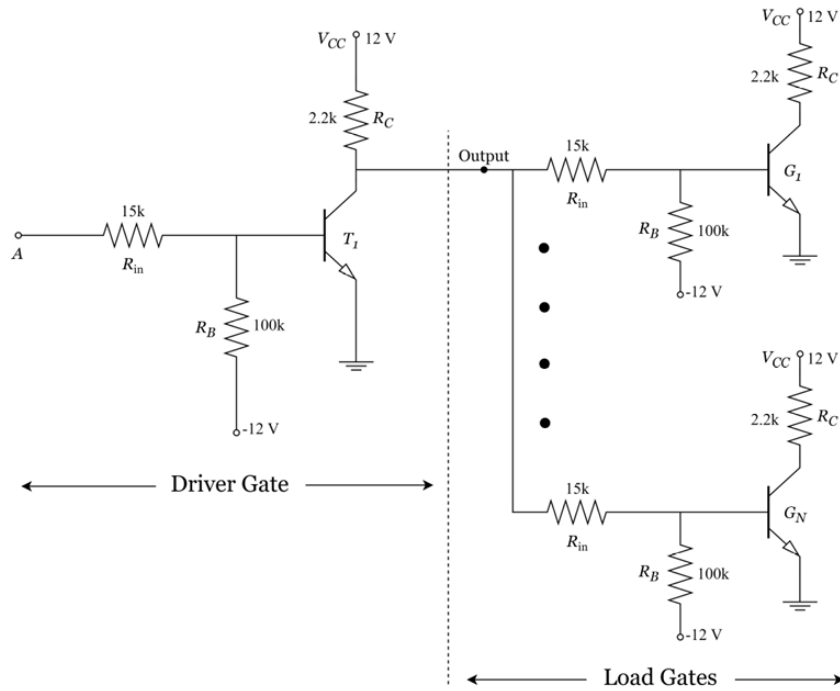
Practice Problem 2:

For the RTL inverter in **Exercise 2**, find the ratio of maximum & minimum power dissipation, if $V_1(-12 V)$ was missing.

Ans: 212494.7:1

Mixed

Exercise 6



For the given RTL inverters,

$$V_{OH} = 10\text{ V}, V_{OL} = 0.2\text{ V}$$

$$\beta_F = 30$$

- Find maximum fanout.
- Find V_O for $N=2$ loads, and the input of the driver is low.
- If V_{in} is high, find the power dissipation of the driver circuit.
- If V_{in} is low and fanout=2, find the power dissipation of the total circuit (driver + loads).
- Find the noise margin.

$$e) 2.175\text{ V}$$

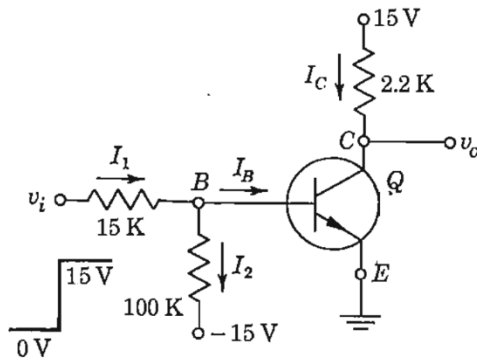
$$d) 146.95\text{ mW}$$

$$c) 73.384\text{ mW}$$

$$b) 9.46\text{ V}$$

$$\text{Ans: a) 1}$$

Exercise 9



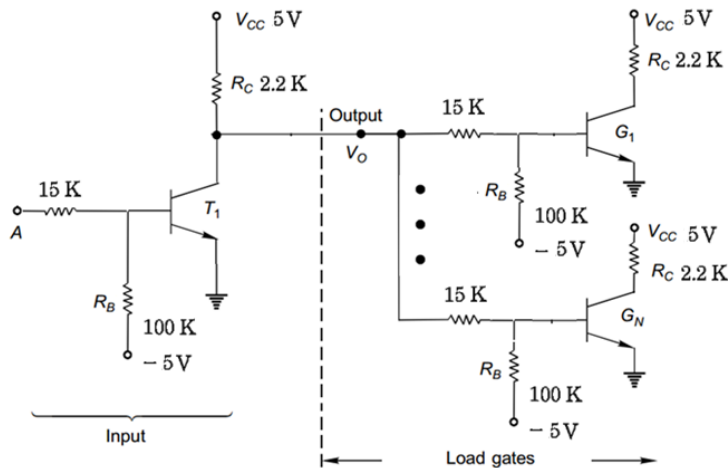
For the RTL NOT gate shown,

When input is high-

- Find the output voltage.
- Find the value of I_C .
- Find the value of I_1 .
- Find the value of I_2 .
- Find the value of I_B .

Ans: a) 0.2 V
b) 6.7272 mA
c) 0.94667 mA
d) 0.158 mA
e) 0.78867 mA

Exercise 10



For the above RTL inverter driver & loads, assume

$$V_{OH} = 4 \text{ V}, V_{OL} = 0.2 \text{ V}, \beta_F = 30$$

- Find the value of V_{IL} .
- Find the value of V_{IH} .
- Calculate the noise margin.
- Find the maximum number of FANOUT for this circuit.

Ans: a) 1.325 V
b) 2.761 V
c) 1.125 V
d) 2